

Master Defense Preparation Document

Project: Cost-Aware Anaemia Prediction in Adult Women **Framework:** SMOTE-CatBoost Ensemble for Non-Invasive Triage

This document contains everything you need to memorize, understand, and defend during your Final Year Project presentation.

Part 1: The Elevator Pitch

(Memorize this to explain your project in 30 seconds) "Our project is a non-invasive, cost-aware public health triage system. We used the NFHS-5 dataset to predict anaemia in rural women using strictly socio-demographic indicators without any clinical blood tests. Because the dataset was severely imbalanced, standard algorithms like Random Forest blindly guessed everyone was anaemic to inflate their accuracy. We stopped this by implementing SMOTE and a CatBoost meta-learner, mathematically neutralizing the bias and creating a deployable Streamlit dashboard for rural ASHA workers."

Part 2: Core Machine Learning Concepts You MUST Know

1. What is "Target Leakage"?

Definition: When a machine learning model is given information during training that it would not realistically possess during a real-world prediction. **How it applies to you:** If you included Hemoglobin (Hb) to predict anaemia, that is target leakage. Hb is the clinical definition of anaemia.

2. What is SMOTE?

Definition: Synthetic Minority Over-sampling Technique. **How it applies to you:** 64.3% of women in Odisha are anaemic. Only 35.7% are healthy. Because the data was so unbalanced, standard models just ignored the healthy people. SMOTE synthetically generated artificial "Healthy" patients based on nearest-neighbor mathematics so the training

set became exactly 50/50. *Note: Explain that you ONLY applied SMOTE to the training data, never the testing data.*

3. Why CatBoost over XGBoost or Random Forest?

Definition: CatBoost (Categorical Boosting) is a specialized gradient-boosting algorithm developed by Yandex. **How it applies to you:** Almost all your 16 features are categorical (Wealth Index: 1-5, Education: 1-4, Eats Meat: 0-2). XGBoost and RF struggle with categorical data and require 'One-Hot Encoding' which expands dimensionality. CatBoost handles categorical data natively using "Ordered Target Statistics," making it vastly superior for survey-based datasets.

4. Accuracy vs AUC

Accuracy: Simply "How many did I guess right?" (Bad metric for imbalanced datasets). **AUC (Area Under the ROC Curve):** Measures the model's ability to distinguish between classes. **How it applies to you:** The Reference Model had 61% accuracy, but an AUC of 0.564. Your model had 53% accuracy, but an AUC of 0.573. You proved that *Accuracy is a lie in medical datasets*, and AUC proves your model actually learned better.

5. Sensitivity vs Specificity

- **Sensitivity (True Positive Rate):** Out of all sick people, how many did you catch?
 - **Specificity (True Negative Rate):** Out of all healthy people, how many did you correctly identify as healthy? **Your tradeoff:** You intentionally sacrificed some Sensitivity to massively increase Specificity (from 26% up to 60%).
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Part 3: Anticipated Questions & Counter-Questions

[!WARNING] **Trap 1: The "Low Accuracy" Trap Professor Asks:** *"Your accuracy is only 53.51%. My other students are getting 95%. Isn't your model terrible?"*

How to defend: "No sir, the groups getting 95% are suffering from Target Leakage or immense class imbalance. If 95% of a dataset has a disease, an AI can achieve 95% accuracy by blindly writing 'Yes' for every patient without learning a single thing. Medical AI

must be judged by **AUC and Specificity**, not accuracy. Our CatBoost model achieved the highest AUC (0.573) by neutralizing algorithmic bias and correctly identifying 60% of healthy patients that baseline models failed to find."

[!WARNING] Trap 2: The "Why exclude Medical Factors?" Trap Professor Asks:
"Why did you throw away Hemoglobin and clinical data? Medicine requires clinical data."

How to defend: "Because we didn't build a laboratory tool, we built a rural triage tool. In remote villages in Odisha, ASHA workers do not have access to needles, cold-storage blood vials, or \$50 to run tests. Our entire research premise was 'Cost-Aware Prediction'. We proved that we can use proxy sociological surveys (wealth, BMI, diets) to successfully identify high-risk women before a clinical test is ever ordered."

[!WARNING] Trap 3: The "Proxy Variable" Trap Professor Asks: *"You seriously used 'Watching TV' and 'Reading Newspapers' to predict a blood disease? That makes no sense."*

How to defend: "In Data Science, these are called 'Socioeconomic Proxy Variables'. Watching TV doesn't cure anaemia. However, machine learning algorithms map multidimensional realities. A woman who watches TV and reads newspapers regularly has a mathematically verifiable higher functional health literacy, a modernized household with electricity (refrigeration for fresh food), and exposure to government Iron-Folic Acid campaigns. The AI learned the sociology, not the biology."

[!WARNING] Trap 4: The "Statistical Significance" Trap Professor Asks: *"How do I know your model being better wasn't just random luck based on how the test set split?"*

How to defend: "We pre-empted that concern by mathematically calculating statistical significance using two distinct tests shown in Table 4 of our paper: 1. **McNemar's Test:** Produced a p-value < 0.0001, proving our model's shift in predictions is fundamentally non-random. 2. **DeLong's Test:** Analyzed the differing ROC curves and returned a p-value of

0.0477 (under the 0.05 index), proving our CatBoost AUC is statistically significantly superior to the baseline."

[!WARNING] **Trap 5: The "SMOTENC" Trap (Very Advanced/Tricky) Professor Asks:**
"Standard SMOTE creates synthetic data via Euclidean distance. Since most of your data is categorical, shouldn't you have used SMOTE-NC (Nominal Continuous)?"

How to defend: "That is an excellent point sir. While SMOTENC is natively built for categorical strings, we first Ordinally Encoded all our categorical data (e.g., Wealth went from 'Poor, Rich' to integers '1, 5'). Because our data exists on an ordered continuum, standard SMOTE interpolation algorithms still functionally mathematically mapped the latent space without producing impossible string classes."

Part 4: Final Tips for the Delivery

1. **Don't Apologize for 53%:** Own it. Medical AI relies on balance, not accuracy. You fixed a mathematical flaw that published researchers fell for.
2. **Push the Streamlit App:** Remind them that yours isn't just a Python script on a Github repo. It is a live GUI dashboard that a doctor or health worker could actually open on an iPad today.
3. **Use the phrase "Cost-Aware":** Keep reminding them that this model costs \$0.00 to run on a patient because it requires zero medical infrastructure.